EDTA, the reaction between EDTA and the metal ions takes place in the 1:1 mole ratio and hence molar solutions are used in this titrations. In the redox and neutralisation titrations we use normal solutions. The mole fraction is used to calculate the partial pressure of gases and the vapour pressure of solutions. The percentage units are used to express the active ingredients present in therapeutics, and the ppm is used to express the quantity of solutes present in small amounts in solutions.

9.3.1 Standard solutions and working standards

A standard solution or a stock solution is a solution whose concentration is accurately known. A standard solution of required concentration can be prepared by dissolving a required amount of a solute, in a suitable amount of solvent. Its done by (i) transforming a known amount of a solute to a standard flask of definite volume. (ii) a small amount of water is added to the flask and shaken well to dissolve the solute. (iii) then water is added to the flask to bring the solution level to the mark indicated at the top end of the flask. (iv) the flask is stoppered and shaken well to make concentration uniform.

At the time of experiment, the solution with required concentration is prepared by diluting the stock solution. This diluted solution is usually called working standard. A known volume of stock solution is transferred to a new container and brought to the calculated volume. The necessary volumes of the stock solution and final volume can be calculated using the following expression.

$$C_s V_s = C_w V_w \quad \text{----------(9.1)}$$

Where the C_s & V_s are concentration and volume of the stock solution and C_w & V_w are concentration and volume of the working standard, respectively.

9.3.2 Advantages of using standard solutions:

1. The error in weighing the solute can be minimised by using concentrated stock solution that requires large quantity of solute.

2. We can prepare working standards of different concentrations by diluting the stock solution, which is more efficient since consistency is maintained.

3. Some of the concentrated solutions are more stable and are less likely to support microbial growth than working standards used in the experiments.

Example Problem 1

1. What volume of 4M HCl and 2M HCl should be mixed to get 500 mL of 2.5 M HCl?

Let the volume of 4M HCl required to prepare 500 mL of 2.5 MHCl = x mL

Therefore, the required volume of 2M HCl = (500 - x) mL

We know from the equation (9.1)

$$C_1V_1 + C_2V_2 = C_3V_3$$

$$(4x)+2(500-x) = 2.5 \times 500$$

$$4x+1000-2x = 1250$$

$$2x = 1250 - 1000$$

$$x = \frac{250}{2}$$

$$= 125 \text{ mL}$$

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